

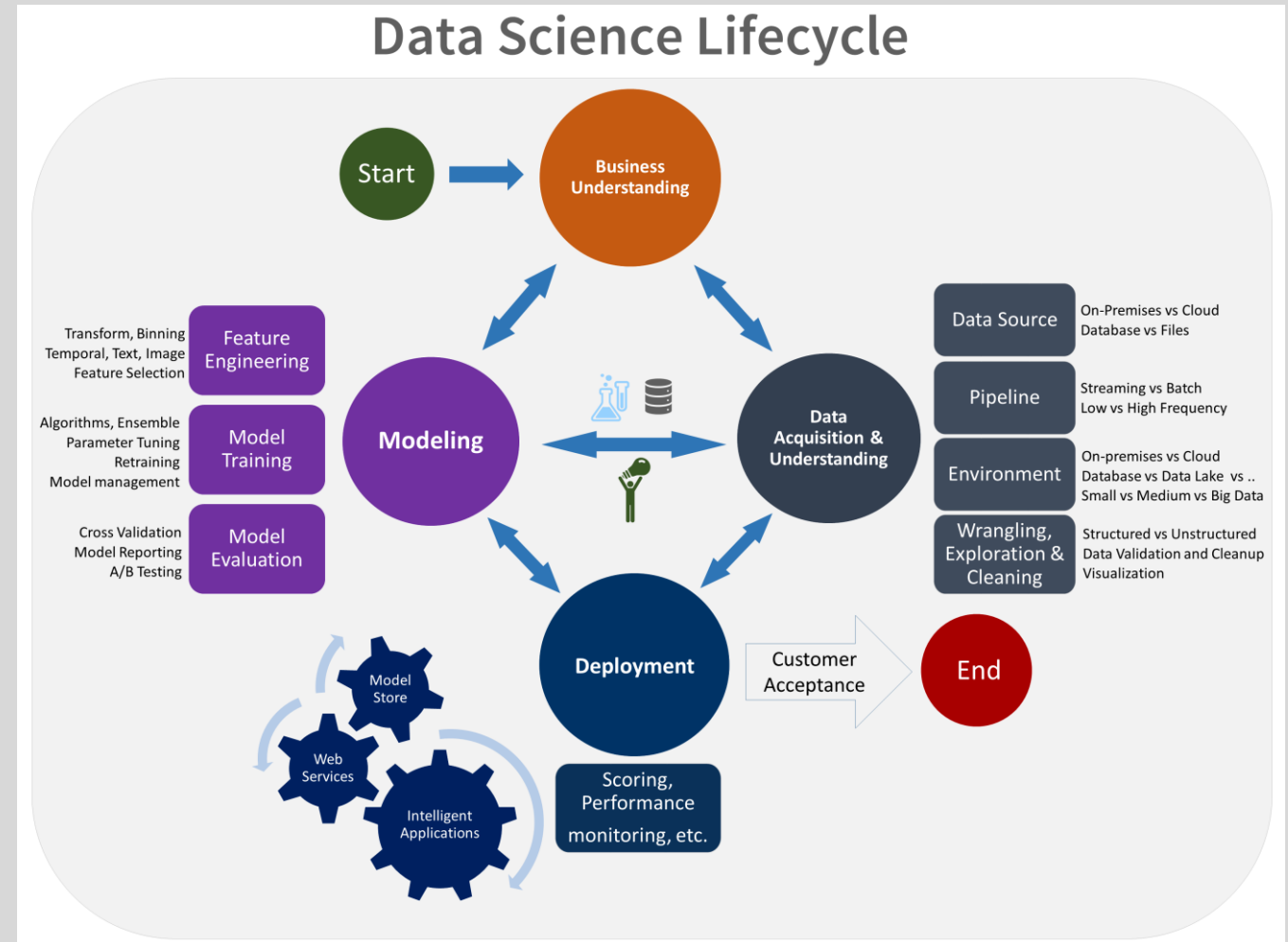


COMPONENTS OF ML PROJECTS

Paul Speaker

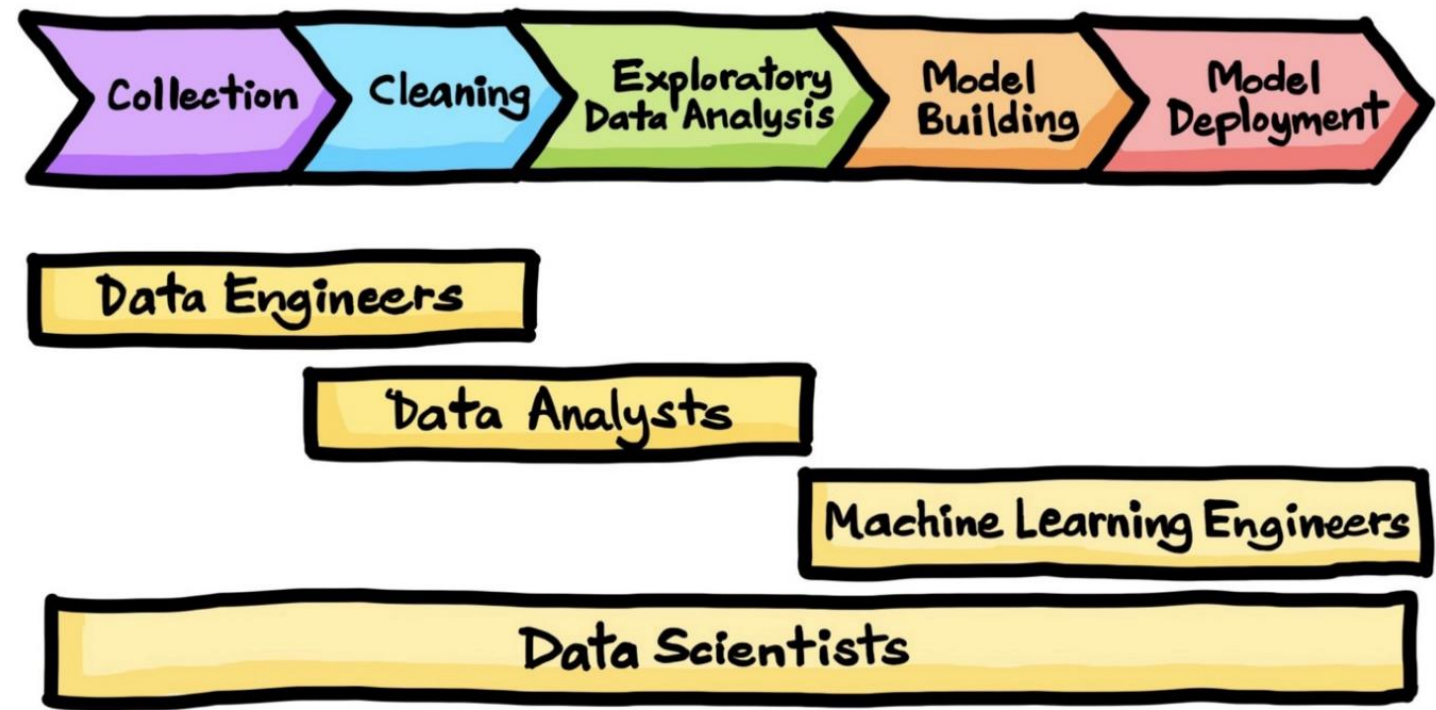
“The” Process

- There are in fact several different representations of what “the” data science process is
- They are mostly saying the same thing
- They also typically leave out many things
- “Process Diagram”
 - Pictorial description of process meant to make supervisors feel good about what you’re doing



Traditional Process Diagram

- What is right with the process shown to the right?
- What is wrong?



Reading Between the Lines

- A process diagram can tell you a lot about an organization
- Who created the process diagram?
- They can tell you
 - Who is running the show
 - What the bottlenecks are
 - How “advanced” a data science organization is
 - Sometimes, they even tell you what data scientists do

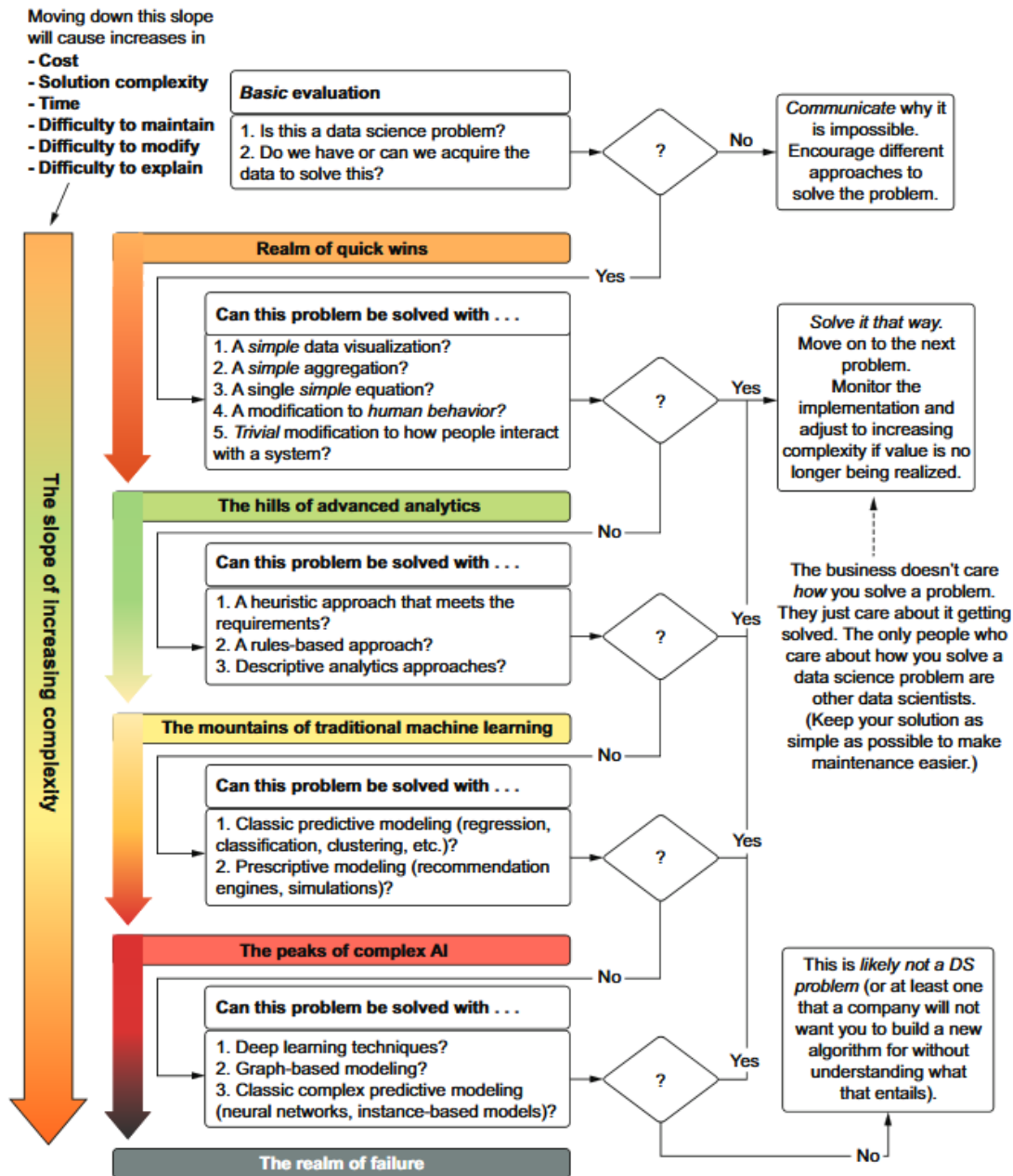


Figure 2.2 Guide for building the simplest solution to an ML problem

Better Process Diagram

Terms to know

- Parsimony—simplest is best
- MVP—minimum viable product
- Versioning/iterating—working with others to improve models
- Agile—process from IT orgs based on “sprints”—improving products in quick bursts

Focus on the Elements

- What goes into a project?
 - The ask
 - The data
 - The model
 - The production
- We covered the first part last time
- We will cover the other parts over the next few weeks
- Next time, we will talk more explicitly about how they are all connected

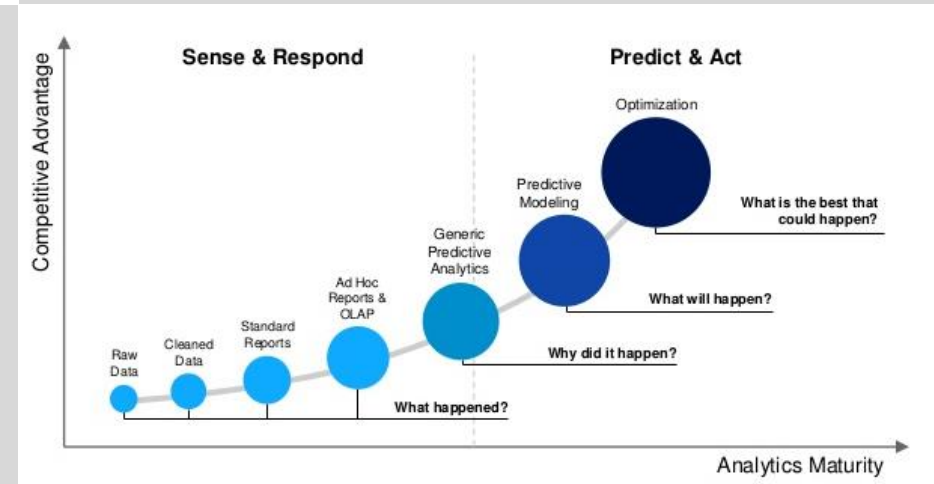
Data Sources

- Project scoping requires a good handle on data needed
- Types of Data
 - Internal vs. External
 - Text vs. Numerical
 - Transactional vs. Summarized
 - Base vs. Change
- How do you get external data?
- How do you get internal data?
- Data for development vs. data for production
 - Death by a 1,000 spreadsheets
- SAP's role in the world

Data Science Org Maturity Levels

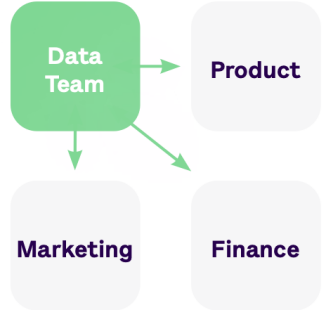
Level 1 Basic	Level 2 Opportunistic	Level 3 Systematic	Level 4 Differentiating	Level 5 Transformational
D&A is transactional and managed in silos Lacks trust in data; analysis is <u>ad hoc</u>	D&A Strategy is not business relevant Lacks leadership support; organizational barriers	Business executives become D&A champions Data types treated differently	Business-led with clear data leadership roles Clear linkages to outcome and business ROI	D&A is central to business strategy Data value influences investments

Gartner Maturity Model for Data and Analytics (D&A)

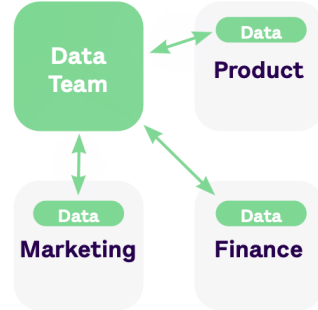


Centralized Analytics Team and “Hub & Spoke”

- 1 Centralized**
Centralized team owns data analytics for the company

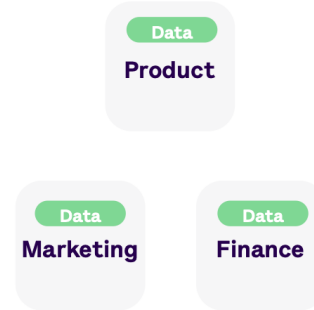


- 2 “Hub & Spoke”**
Central team and distributed data analysts share ownership



Decentralized

- 3 Decentralized**
Creates data silos and can result in inefficiencies, but each team moves fast on their own analytics needs



Org Charts

An Org chart is a graph of how one organization fits within the larger company

Where are data scientists?
3 models

git in Data Science

- Git repositories are a basic tool for many data science organizations
- The more people that work on a project, the more essential git is for data science
- Why?
 - Version control
 - Track changes
 - Security settings
 - Imposes organization

Some terms in git

- **Repository** - "Database" of all the branches and commits of a single project
- **Branch** - Alternative state or line of development for a repository.
- **Merge** - Merging two (or more) branches into a single branch, single truth.
- **Main / Master** - Common name for the root branch, which is the central source of truth.
- **Stage** - Choosing which files will be part of the new commit
- **Commit** - A saved snapshot of staged changes made to the file(s) in the repository.
- **HEAD** - Shorthand for the current commit your local repository is currently on.
- **Push** - Pushing means sending your changes to the remote repository for everyone to see
- **Pull** - Pulling means getting everybody else's changes to your local repository
- **Pull Request** - Mechanism to review & approve your changes before merging to main/master